

FUTURE PROSPECTS FOR IMMUNIZATION IN THE AMERICAS

Twenty-five years ago, under the umbrella of the Expanded Program on Immunization, immunization programs in the countries of the Americas were challenged to rapidly accelerate and expand services. At that time, vaccines for only six diseases were included in the strategy to protect children against vaccine-preventable diseases: measles, pertussis, tetanus, diphtheria, polio, and tuberculosis. Of these six diseases, the leading killers were measles, pertussis, and tetanus. The most crippling disease was polio. The cost to protect against these diseases was less than US\$ 1.00 per child. Today, PAHO's Regional immunization program delivers vaccines against four other diseases that are major causes of morbidity and mortality among children: rubella, hepatitis B, *Haemophilus influenzae* type b, and mumps. While the cost of protecting each child has increased to approximately US\$ 24, countries remain highly committed to sustaining these advances in prevention by purchasing nearly all their vaccines with funds from their own national government budgets.

The major challenge for immunization programs in developing countries is to reduce existing inequities while ensuring that expanded services are sustainable. Program sustainability is best ensured when governments themselves cover the costs of their programs, especially the purchase of new vaccines, instead of turning to external sources of funds. While taking on new immunization activities, efforts must continue to promote better utilization of the vaccines against "old" killer diseases, such as pertussis and tetanus. Whatever the threat is, the vaccine to tackle it must be safe, effective, and affordable. Price remains a vital concern for countries, and it must be a key element to be addressed in PAHO's support to countries. Because safety is also a critical issue, the PAHO Revolving Fund only purchases vaccines that are prequalified by the World Health Organization. The Revolving Fund provides an effective procurement and financing mechanism that complements a broad spectrum of technical assistance. To that end, the Revolving Fund will remain the center piece in PAHO's efforts to support the introduction of new and underutilized vaccines and to ensure their affordability to all, including the children and families who need them the most.

The collaboration between the Regional immunization program and all the national programs in the Americas has resulted in achieving such milestones as the eradication of polio and the elimination of indigenous measles from the Western Hemisphere. However, the elimination of measles

unveiled the "tip of the iceberg" for the disease burden caused by rubella virus. Rubella maims children by causing birth defects to babies born from infected mothers. These birth defects include hearing loss, cataracts, heart abnormalities, and mental retardation. To deal with that, in 2003 the countries of the Americas joined forces to undertake a rubella and congenital rubella syndrome elimination initiative (7).

The initiative to eliminate rubella and congenital rubella syndrome will challenge national immunization programs to reach adults (8). Indeed, childhood immunization programs are rapidly evolving into family immunization programs. The lessons learned by vaccinating both men and women with vaccines containing rubella should pave the way for when vaccines against human papilloma virus and HIV become available. In addition, the rubella elimination initiative will be called on to help improve women's overall health (9, 10).

The achievements of the PAHO Regional immunization program are far from sufficient. Vaccines against "old" diseases, such as pertussis and tetanus, still need to be delivered more effectively. Pertussis continues to kill approximately 12 000 children per year in the Americas. PAHO's approach to managing its Regional immunization program, developed over the last 25 years, needs to continue efforts to address issues of inequity and to ensure sustainability. Safe, effective, affordable new vaccines, such as those for rotavirus and pneumococcus infections, must be made available to the people who need them most, including persons living in poor and difficult-to-access communities.

LESSONS LEARNED

The PAHO Regional program on immunization has built institutional structure at the national and international levels. Fundamental to this process has been that the national immunization programs are supported by Regional bodies that provide technical oversight, partnership and coordination, surveillance, networks of public health laboratories, and vaccine purchase through the PAHO Revolving Fund.

Based on the lessons learned from the years of experience coordinating immunization activities among all the countries in the Americas, other regional and global health programs should be grounded in the following characteristics:

- A core value of cooperation among countries. This was the experience of Pan-Americanism.
- The capacity to identify problems and design appropriate solutions. EPI tackled abysmally low